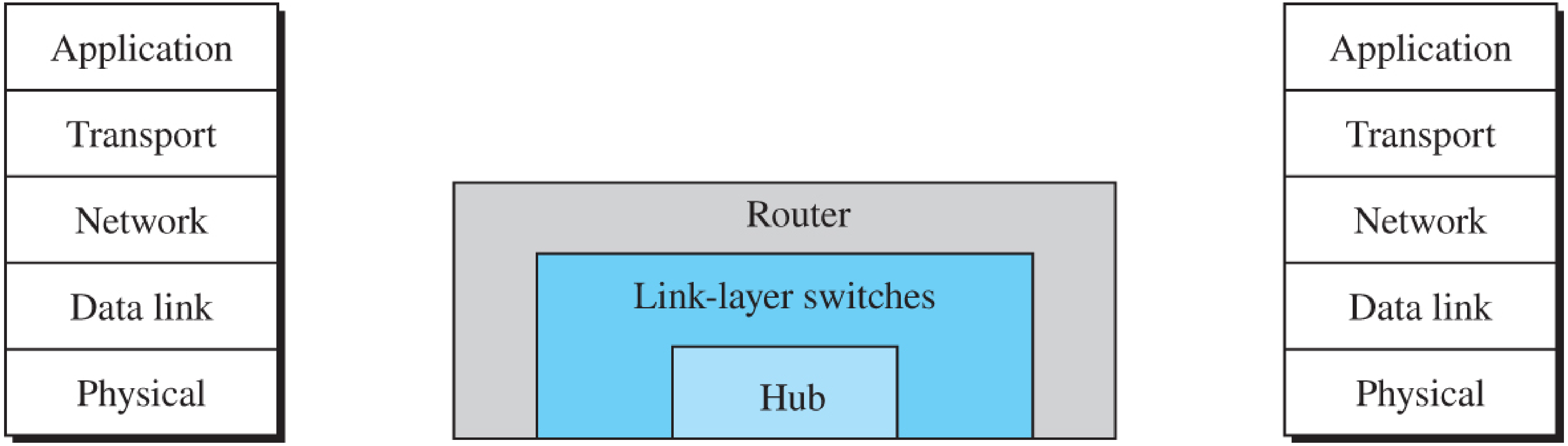
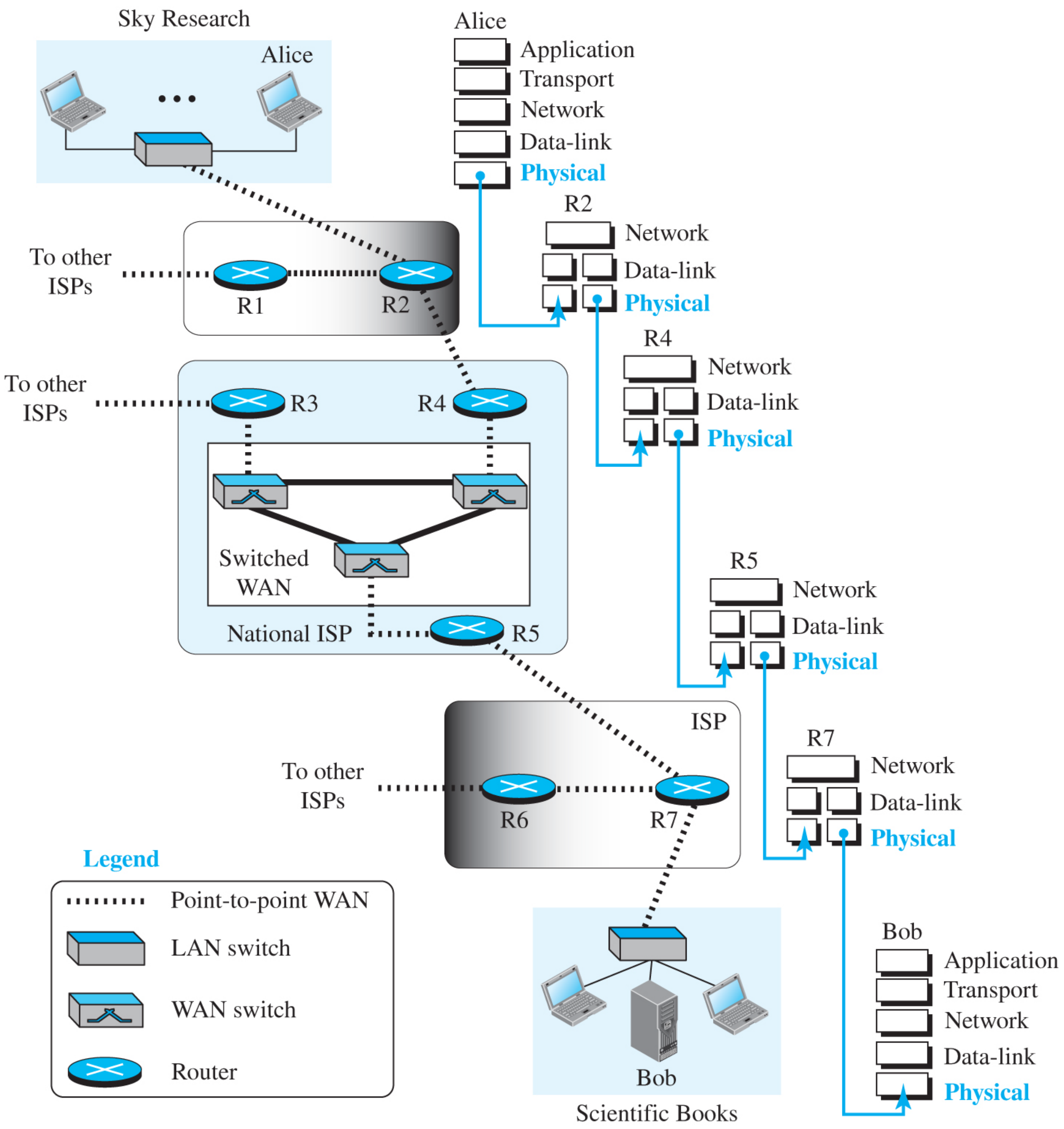
EGCI 371
Computer Network

Networking Devices

Connecting Devices & VLANs

- Hubs, link-layer switches, routers
- Why VLANs exist; how they're implemented
- Learning goals: eliminate collisions, forward efficiently, segment logically





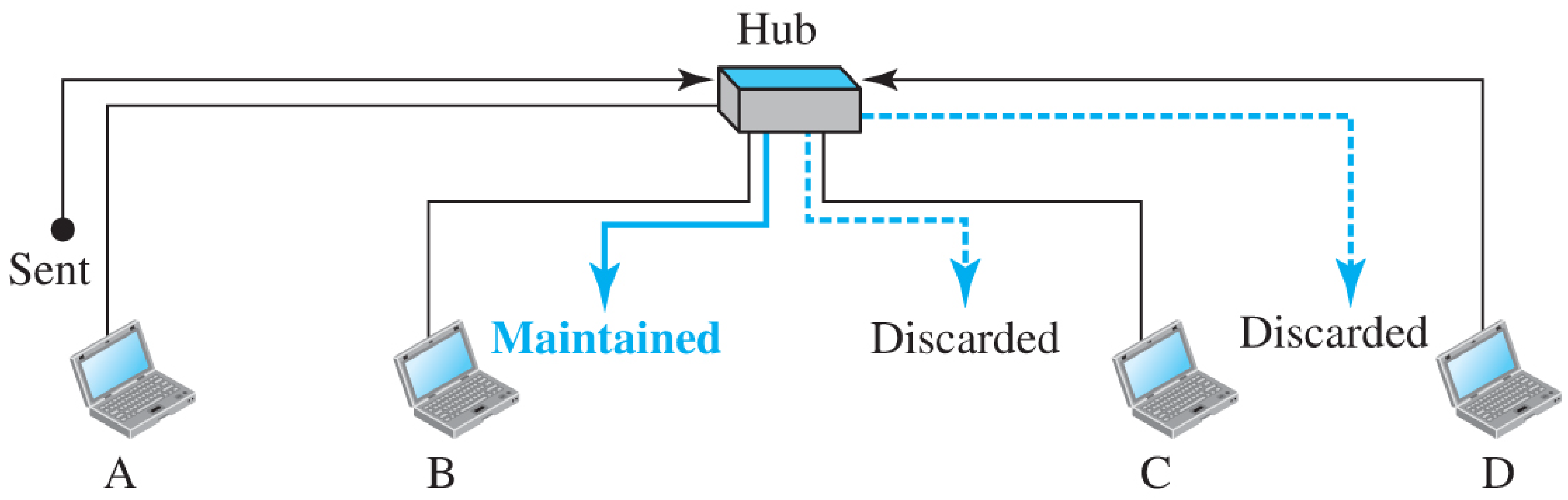
Sky Research
Alice
To other ISPs
R1
R2
To other ISPs
R3
R4
Switched WAN
National ISP
R5
ISP
To other ISPs
R6
R7
Bob
Scientific Books
Legend
Point-to-point WAN
LAN switch
WAN switch
Router
Alice
Application
Transport
Network
Data-link
Physical
R2
Network
Data-link
Physical
R4
Network
Data-link
Physical
R5
Network
Data-link
Physical
R7
Network
Data-link
Physical
Bob
Application
Transport
Network
Data-link
Physical

Where Devices Live (Layers)

- Hubs: Physical layer repeat/regenerate bits
- Switches: Physical + Data Link (inspect MAC addresses)
- Routers: Physical + Data Link + Network (use IP/routing)

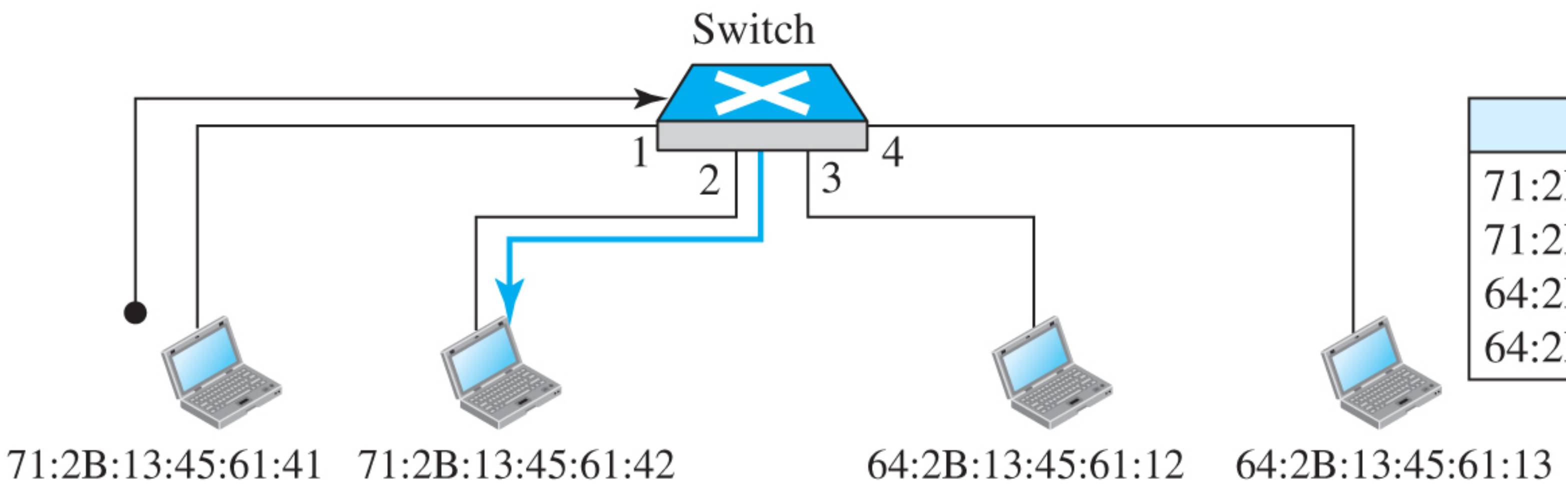
Hubs (Repeaters)

- Extend reach: regenerate & retime before signal degrades
- One shared collision domain → broadcasts to all ports
- Legacy; useful to contrast with switches



Link-Layer Switches (What They Add)

- Check source/destination MAC in frames
- Decide which port to forward out (filtering)
- Regenerate signal + isolate ports (micro-segmentation)



Switching table

Address	Port
71:2B:13:45:61:41	1
71:2B:13:45:61:42	2
64:2B:13:45:61:12	3
64:2B:13:45:61:13	4

Filtering vs Flooding

- Known dest MAC → unicast out a single port
- Unknown dest MAC/broadcast → flood out all except ingress

Gradual building of table

Address	Port

a. Original

Address	Port
71:2B:13:45:61:41	1

b. After A sends a frame to D

Address	Port
71:2B:13:45:61:41	1
64:2B:13:45:61:13	4

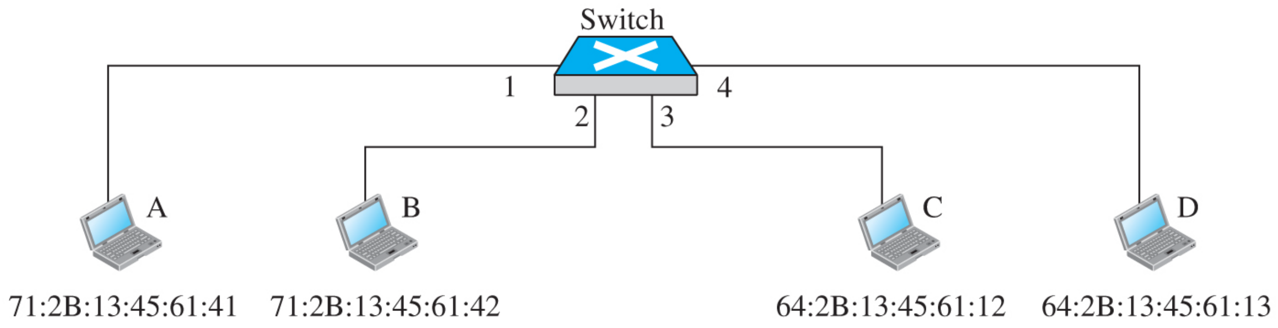
c. After D sends a frame to B

Address	Port
71:2B:13:45:61:41	1
64:2B:13:45:61:13	4
71:2B:13:45:61:42	2

d. After B sends a frame to A

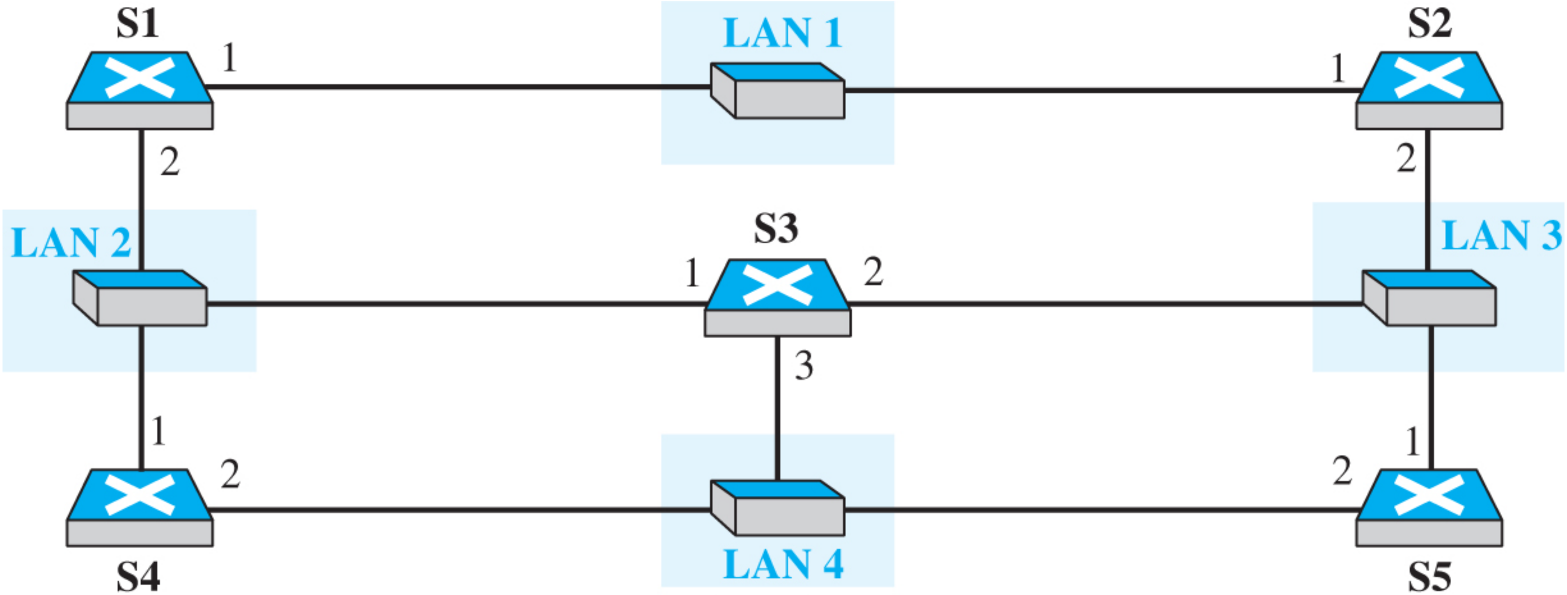
Address	Port
71:2B:13:45:61:41	1
64:2B:13:45:61:13	4
71:2B:13:45:61:42	2
64:2B:13:45:61:12	3

e. After C sends a frame to D

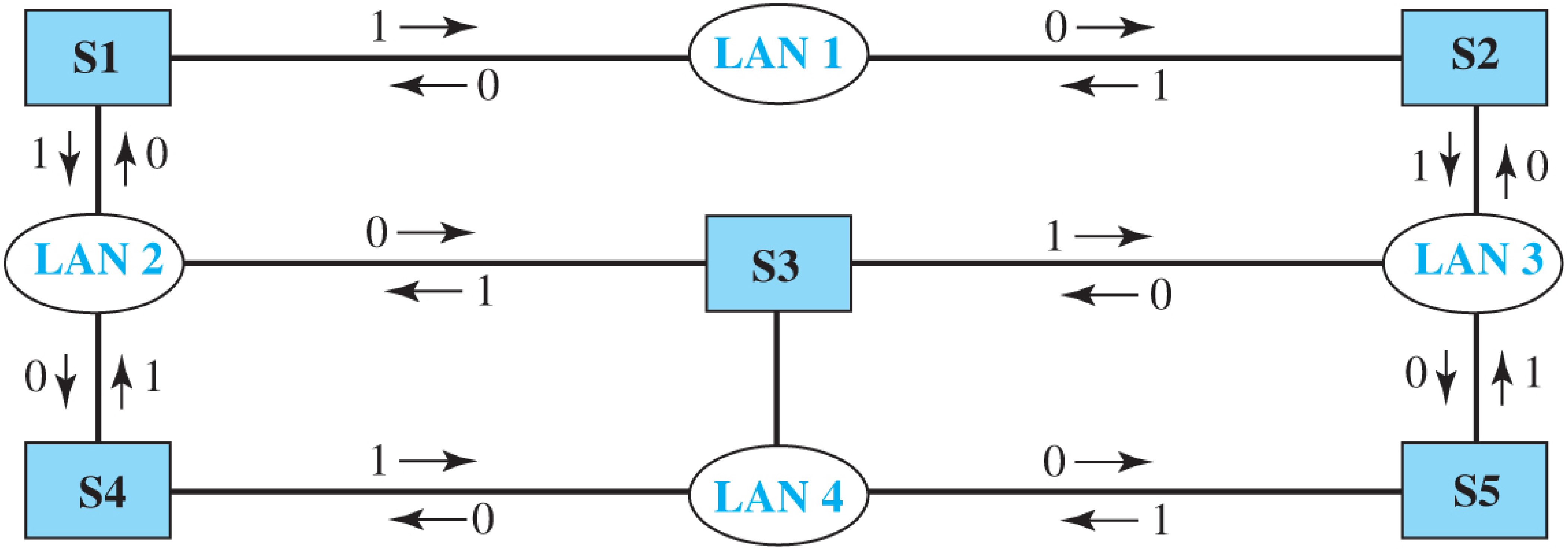


Loops Break Learning (Why STP)

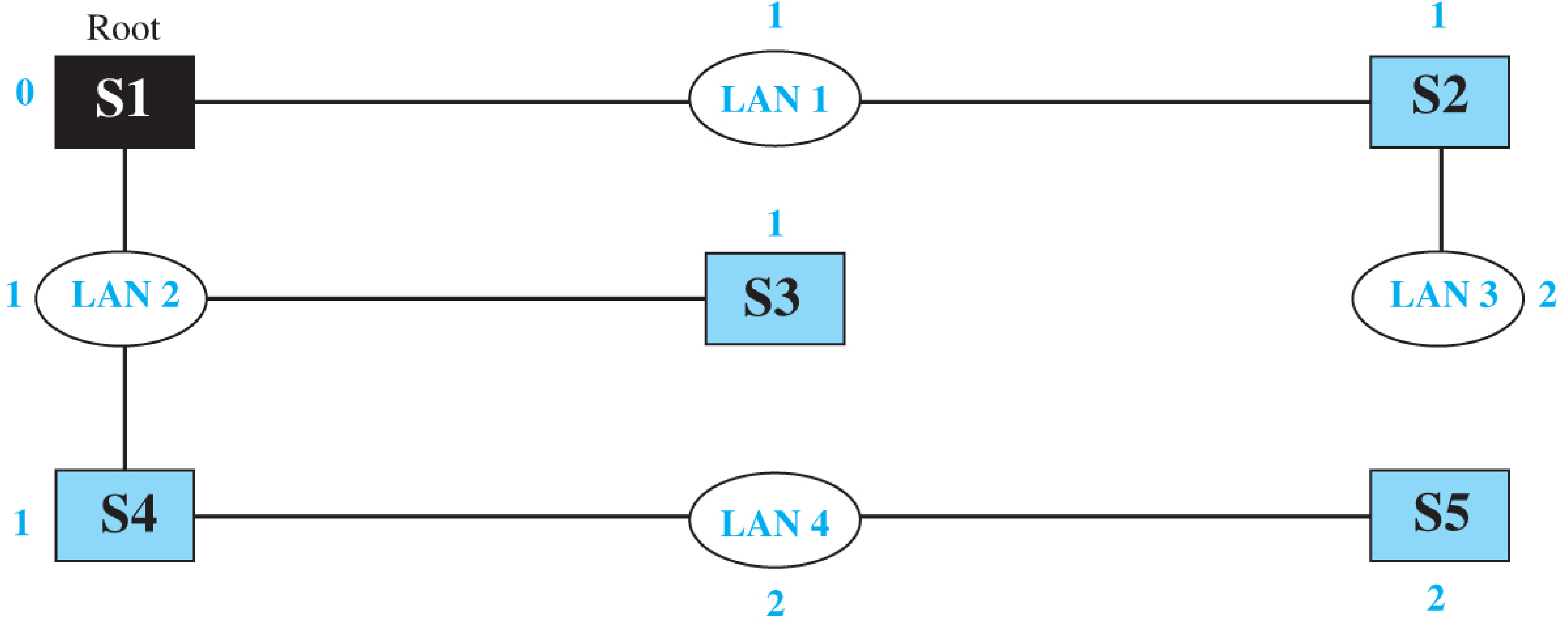
- Redundant links create loops → repeated flooding, table instability
- Need a loop-prevention protocol → spanning tree blocks some ports
- Result: loop-free tree across the LAN



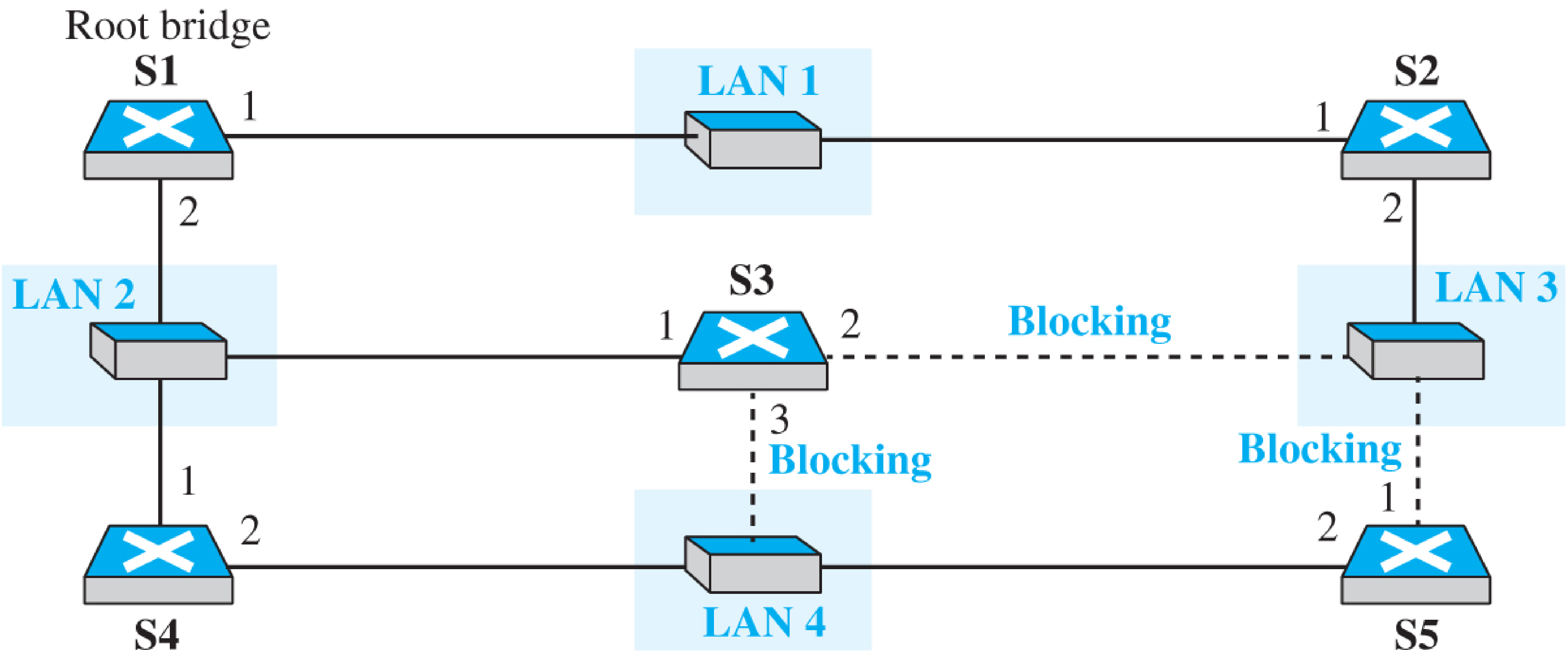
a. Actual system



b. Graph representation with cost assigned to each arc


Root
0
S1
1
LAN 1
1
S2
1
1
LAN 2
S3
LAN 3
2
1
S4
LAN 4
S5
2
2

Ports 2 and 3 of bridge S3 are blocking ports (no frame is sent out of these ports).
Port 1 of bridge S5 is also a blocking port (no frame is sent out of this port).



Spanning Tree Outcomes

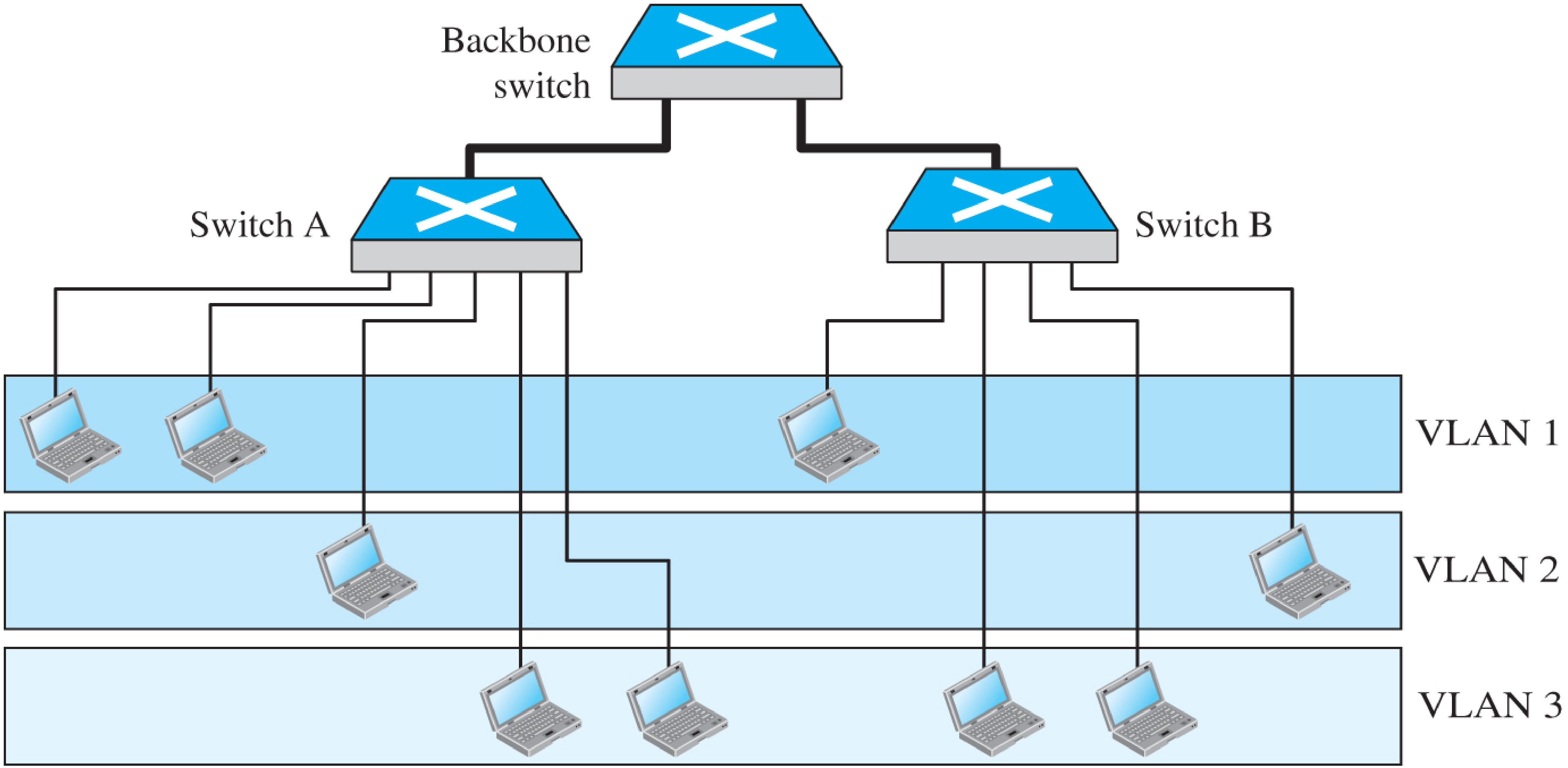
- Compute shortest path; forwarding vs blocking ports
- Converged topology avoids broadcast storms
- Restores stable learning/forwarding

VLANs — Why “Virtual” LANs?

- Group stations by software, not geography/cabling
- Same switch can host multiple logical LANs
- Reduces physical rewire; aligns network with org structure
- By port/interface, MAC, IP, multicast, or combinations
- Admin chooses one/more attributes during setup
- Flexible grouping across floors/buildings

VLAN Advantages (So What?)

- Cost/time reduction: re-segment in software, not cabling
- Virtual workgroups: project teams can broadcast within group
- Security: group broadcasts don’t leak to other VLANs

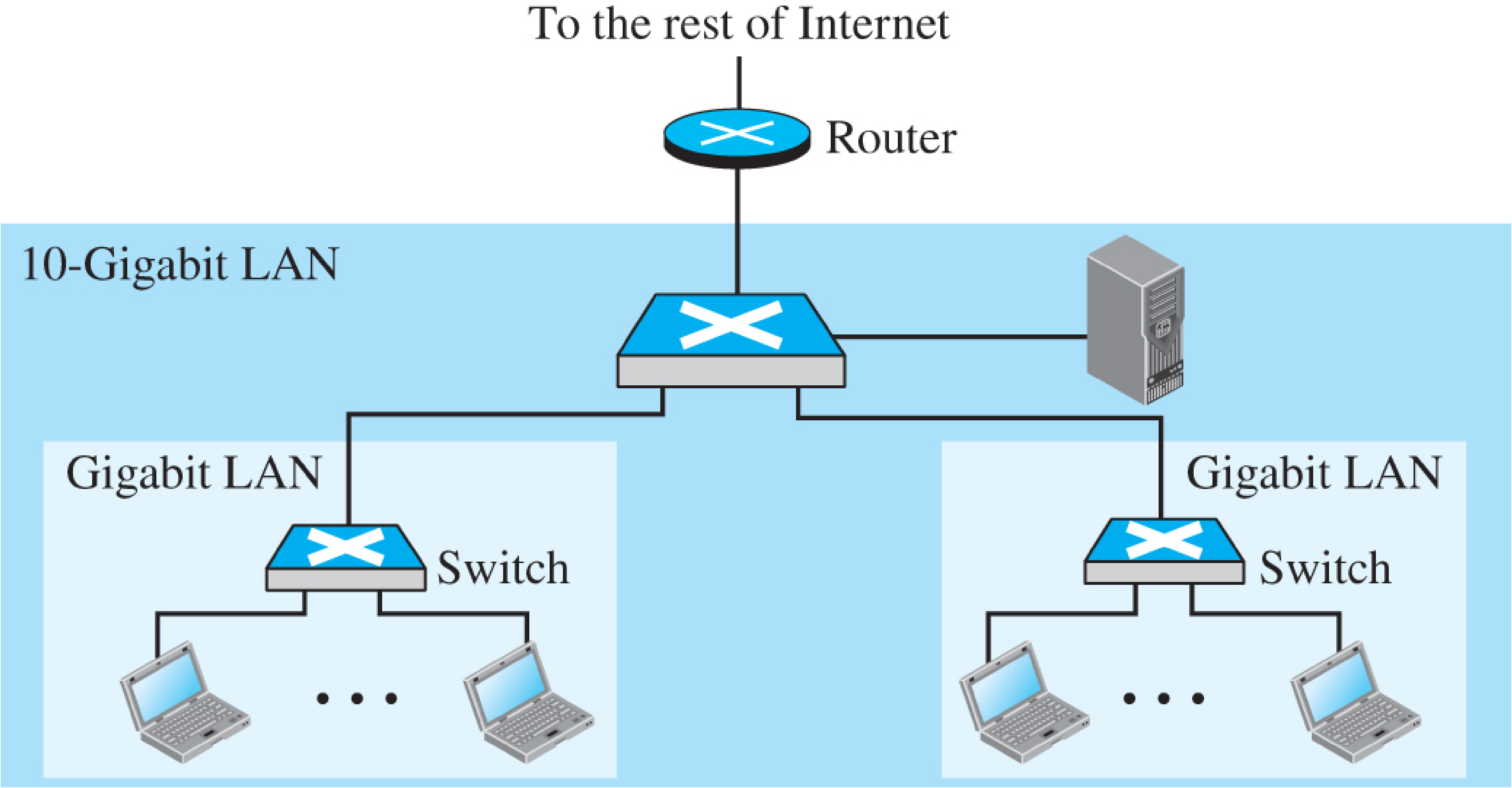

Backbone switch
Switch A
Switch B
VLAN 1
VLAN 2
VLAN 3

Frame Tagging (802.1Q Focus)

- Add an extra tag in the MAC frame when crossing trunks
- Receiving switch reads tag → knows target VLAN
- IEEE 802.1Q: the open standard enabling multivendor VLANs

Routers (Quick Contrast)

- Make L3 decisions using IP and routing tables
- Separate broadcast domains; require ARP per segment
- Path selection across networks (discussed deeper in routing chapters)


To the rest of Internet
Router
10-Gigabit LAN
Gigabit LAN
Switch
Gigabit LAN
Switch